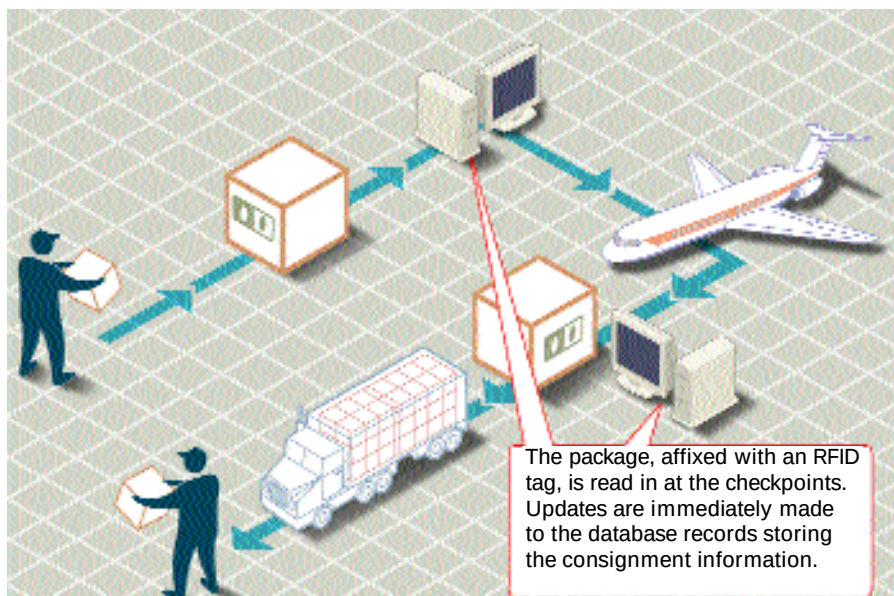




endurance of the onboard battery. They are also more expensive to produce, and are larger.

Readers and tags can use low frequencies—30 KHz to 500 KHz. Low frequencies means lower ranges—up to a few metres. They may use high-frequency communication systems too, such as those in the 2.4 GHz range. These offer far longer ranges—up to 25 metres. However, they're more expensive to build and operate.

RFID in action

The market for RFID solutions is huge. RFIDs can network just about anything—shipping containers, clothes, books and even pets! The manufacturing and retail logistics sectors are big users of current UPC technologies. So are the service sectors that move shipments across the world. RFID would enable them to monitor production and eliminate wastage, since inventory and other data would be real-time. In fact, Wal-Mart has mandated that their top suppliers start using RFID technology, and that all shipping cases be RFID-enabled, come 2005. The US Department of Defence is also considering mandating RFID.

The organised Indian retail sector would be interested in RFID too. So would the Indian transportation logistics sector. S. R. Mallela, Chief Technology Officer of AFL, an express parcel delivery service, says,

“RFID, coupled with other technologies, will help us track our shipments from point to point”. He should know: AFL is a major

player in the Indian logistics services sector, as a leading parcel delivery service.

AFL is considering implementing RFID, coupled with other tracking systems based on Global Positioning System (GPS) or General Packet Radio Service (GPRS), which will help deliver up-to-date information to customers about their shipments.

Despite the substantial investment needed to implement the technology, Mallela points out that the returns in terms of customer benefit make it worth it. Currently, information while tracking a shipment is obtained only when it reaches a checkpoint in the system; an RFID system in conjunction with real-time tracking using GPS, would let customers track their goods on a minute-to-minute basis over the Internet. While AFL is not using RFID-based tracking systems now, they intend to conduct feasibility trials over the first half of this year.

Other RFID applications include tracking farm and wild animals. Animal conservation experts in several places—Africa, for example—already use RFID microchips to track animals habits and to ensure their well being. For example, if an animal is in the same place for a long time, it might mean that it's been trapped.

In a pilot project, the Indian Ministry of Environment and Forests worked on

getting RFID microchips implanted in elephants in north-eastern India in 2002. The idea was to create



Privacy

Wired Magazine reported early this year that Intel and the Georgia Institute of Technology were working on systems that kept track of senior citizens and their health at any point in time, using RFID. Sounds like an amazing application, but there's a trade-off with privacy. Pervasive RFID could well let Big Brother keep tabs on you. Of course, in the case of Intel and GeorgiaTech, the seniors signed into the program, and the data went only to caregivers who monitored the seniors' routine for any indications of illness.

Imagine RFID-enabled clothes that could keep tabs of your movements and preferences. Worse still, think about RFID tags that could keep tabs of who you interact with, when and where. Most of this, admittedly, is doomsday fantasy material, but deep privacy concerns do exist: if anything can be tagged, and if that tag could keep reading in all sorts of information and potentially spill it out to those who shouldn't have it, privacy is threatened. RFID technology implementation in the mainstream will have to address that before everyone's happy.

and maintain a database of the elephants and track their habits.

Software giant Patni Computer Systems worked on a pilot of an animal tracking system, using RFID, for a major US-based client, which, in Dilip Dhanuka's words, “tracks the animal from farm to plate”. Dhanuka, General Manager of Products and Technology group at Patni Computer Systems, says that the system works using RFID readers and a desktop synchronisation device—a service that allows one to view and control everything from one central location. A Web application allows users to aggregate, synchronise and view data across multiple farms and the abattoir, and also to modify the database. The readers, also called

‘Romeos’—‘Juliet’ is the desktop synchronisation device—can be interfaced with other external devices, such as thermometer sensors.

RFID technology can be employed in security and identification systems as well. In an interesting case, the European Central Bank is